

Genetic report for longevity

Hair

Androgenetic alopecia, white hair, thickness and shape, alopecia areata, nutrients and pharmacogenetics of treatments

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The hair follicle is a mini-organ that regenerates at each cycle; hair loss and white hair are among the most visible signs of aging of stem cells. This report reads the genes of androgenetic alopecia (androgen receptor in chromosome X, 5-alpha-reductase, autosomal loci), early white hair, hair thickness and shape, alopecia areata and nutrients, and pharmacogenetics of treatments.

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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Summary

31

variants

25

in the chip

4

imputed
($DR2 \geq 0.8$)

2

not found

4

Scores with
percentile

Overall reading. The genetic risk of male pattern baldness is average: the score of 365 autosomal loci is at the 32nd percentile and the PGS of the pattern of baldness at the 59th percentile, and in the chromosome X (gene AR, which explains a large part of heredity) has the common allele of risk G, as about 86% of men in the reference population. Mean, in men, still means high absolute risk (about half has some degree of baldness at the age of 50), so surveillance by photographs and an early treatment, if there is evolution, are worth more than genes. In favor: HDAC9, TWIST2 and IRF4 (rs9392026) without risk alleles; against: FAM53B CC. In the alopecia areata, the score is at the 28th percentile (small loci; the HLA region was not read) and the CTLA4 is favourable; the PGS of hair and follicle diseases (large clinical category) is at the 79th percentile, unfavourable but not specific. In white hair, there is no IRF4 allele linked to early greying. In iron, the heterozygous HFE H63D gives a slight tendency to accumulate, which reinforces the target of the plan (ferritin <100); biotinidase (BTD) is normal.

Favourable

- **HFE rs1800562 GG:** C282Y: A allele is the main hereditary hemochromatosis (AA = iron overload).
- **HDAC9 rs67248060 CC:** HDAC9 (7p21).
- **TWIST2 rs11684254 CC:** TWIST2: follicle development.
- **IRF4 rs9392026 AA:** IRF4: also baldness locus.
- **CTLA4 rs231775 AA:** CTLA4: T lymphocyte brake.

Unfavourable

- **Hair and hair follicle diseases:** the 78th percentile
- **HFE rs1799945 CG:** H63D: the G allele slightly increases iron absorption; the plan already fixed ferritin below 100.
- **FAM53B rs3781452 CC:** FAM53B.

Here, the scores out of the typical range (≤ 25 or ≥ 75 , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 25 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 1 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

Androgenetic alopecia (male pattern baldness)

Androgenetic alopecia is very hereditary (~80%). The strongest locus is that of the androgen receptor (AR/EDA2R, X chromosome, inherited from the mother); followed by dozens of autosomal loci (20p11 PAX1/FOXA2, HDAC9, EBF1, TWIST2, SRD5A2 and others). The score of pattern of baldness (PGS001987) and the score of loci of GWAS use only the self-somes; the chromosome X is read apart.

Ageing mechanisms: Exhaustion of Stem Cells; Endocrine dysregulation; Cell Senescence; Chronic Inflammation

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Baldness pattern (autosomes)	Typical / intermediate	the 59th percentile · z=0.19	Higher score = more baldness. Without the X chromosome (AR), which is read separately. PGS001987 (PGS Catalog; Penalized regression; Hair/balding pattern; PGP000263, PMID 34995502). Variants with genotype: 22944 of 23692 (covering of 97.2% coefficients).
Male pattern calvisys (GWAS loci, autosomes)	Typical / intermediate	the 32nd percentile · z=-0.45	Higher score : higher risk. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; ln(OR) or beta coefficients): 354 of 365 loci with genotype. Studies: Hagenaars SP (Male-pattern range); Heilmann-Heimbach S (Male-pattern range); Yap CX (Male-pattern range). Variants with genotype: 354 of 365 (covering of 91.2% coefficients).

Favourable

- HDAC9 rs67248060 CC
- TWIST2 rs11684254 CC
- IRF4 rs9392026 AA

Unfavourable

- FAM53B rs3781452 CC

Favourable variants (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HDAC9	rs67248060 · 7:18859540 · C/A/T	CC	chip	T (0; 38%)	moderate	HDAC9 (7p21). Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=3e-283); T ↑ allele
TWIST2	rs11684254 · 2:238787252 · C/G/T	CC	chip	G (0; 35%)	moderate	TWIST2: follicle development. Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=1e-146); G ↑ allele
IRF4	rs9392026 · 6:453415 · G/A	AA	chip	G (0; 34%)	moderate	IRF4: also baldness locus. Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=2e-64); G ↑ allele

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
FAM53B	rs3781452 · 10:124666560 · C/T	CC	chip	C (2; 40%)	moderate	FAM53B. Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=5e-89); C ↑ allele

Typical / intermediate variants (7)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
AR		G	chip	G (1; 86%)	moderate	E211 (synonymous, 1 exon of androgen receptor, X chromosome): G allele is the risk of androgenetic alopecia (Hillmer

	rs6152 X:67545785 · G/A/T					2005); as men have 1 X, it counts 1 copy.
AR	rs2497911 X:67407598 · C/A	C	chip	C (1; 82%)	moderate	Androgen receptor region (chromosome X): the strongest locus in the baldness. Meaning: GWAS Catalog: Male-pattern range (Heilmann-Heimbach S 2017; p=1e-320); C ↑ allele
OPHN1	rs1475417 X:68177797 · G/A	G	imputation DR2=1.00	A (0; 12%)	moderate	EDA2R/OPHN1 region (X chromosome), neighboring the AR. Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=4e-178); Allele A ↑
LINC01432	rs6113492 20:22077868 · G/A/C/T	CG	imputation DR2=1.00	G (1; 57%)	moderate	Locus 20p11 (PAX1/FOXA2): the second strongest locus. Meaning: GWAS Catalog: Male-pattern range (Heilmann-Heimbach S 2017; p=1e-162); G ↑ allele
EBF1	rs7736883 5:158916961 · A/G	GA	imputation DR2=0.99	A (1; 64%)	moderate	EBF1: transcription factor; baldness locus. Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=4e-267); G ↓ allele
SRD5A2	rs76712439 2:31689852 · C/T	CC	imputation DR2=1.00	C (2; 97%)	moderate	5-alpha-reductase 2 region (main variant of GWAS). Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=9e-99); T ↓ allele
RUNX3	rs188468174 1:24965206 · C/T	CC	chip	C (2; 100%)	moderate	RUNX3. Meaning: GWAS Catalog: Male-pattern range (Yap CX 2018; p=6e-93); T ↓ allele

Non-judgmental variants (informative) (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SRD5A2	rs523349 2:31580636 · G/A/C/T	CC	chip	—	weak	V89L (5-alpha-reductase type 2): the allele 89L (G in strand +) lowers the activity of the enzyme ~30%; association with weak baldness. Informative for finasteride.
SRD5A2	rs9282858 2:31580756 · C/T	CC	chip	—	weak	A49T: rare variant that increases the activity of 5-alpha-reductase type 2. Information.

Early white hair

White hair results from depletion of stem cells in the follicle and oxidative stress (H2O2). IRF4 is the only well replicated locus for the beginning of white hair.

Ageing mechanisms: Exhaustion of Stem Cells; Oxidative stress; Cell Senescence

Favourable

None.

Unfavourable

None.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
IRF4	rs12203592 6:396321 · C/ G/T	CC	chip	T (0; 12%)	moderate	The T allele is associated with earlier white hair (Adhikari 2016), freckles and sunspots.

Hair thickness, shape and density

Genes of the shape and thickness of the wire (TCHH, EDAR, WNT10A). They are traits without good/bad judgment; they inform about the visual "reservoir" of the hair.

Ageing mechanisms: Exhaustion of Stem Cells

Favourable

None.

Unfavourable

None.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TCHH	rs11803731 1:152110849 · A/C/G/T	AT	chip	—	moderate	Trichohialine (L790M): the T allele is associated with smoother hair; the A allele is associated with more wavy hair. No judgment.
EDAR	rs3827760 2:108897145 · A/C/G	AA	chip	—	strong	V370A: the G allele (370A) gives thicker thread and smoother hair; it is rare in the reference population. No judgment.
WNT10A	rs7349332 2:218891661 · C/G/T	CC	chip	—	weak	WNT10A: loci of shape and thickness of the hair and the barb. Information.

Alopecia areata and autoimmunity of the follicle

In alopecia areata, T lymphocytes attack the follicle when it loses "immune privilege". Loci: HLA-DQB1, ULBP/RAET1 (NKG2D binders), IL2RA, CTLA4, IKZF4, PRDX5. Also enters the score of hair and follicle diseases (PGS001873).

Ageing mechanisms: Chronic Inflammation; Immunosenescence; Intercellular Communication

Polygenic and loci scores

Score	Result	Percentage (reference population)	Reading and method
Hair and hair follicle diseases	Unfavourable	the 78th percentile · z=0.63	Higher score = more risk (clinical category that joins alopecia, folliculitis and others). PGS001873 (PGS Catalog; Penalized regression; Diseases of hair and hair follicles; PGP000263, PMID 34995502). Variants with genotype: 453 of 461 (covering of 99.2% coefficients).
Alopecia areata	Typical / intermediate	the 28th percentile · z=-0.62	Higher score : higher risk. Loci score of GWAS Catalog (p≤510–8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 10 of 22 loci with genotype. Studies: Betz RC (Alopecia areata); Verma A (Alopecia areata (PheCode 704.11)); Yang JS (Alopecia areata). Variants with genotype: 10 of 22 (covering of 45.5% coefficients).

Favourable

- CTLA4 rs231775 AA

Unfavourable

- Hair and hair follicle diseases (the 78th percentile)

Favourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
CTLA4	rs231775 2:203867991 · A/G/T	AA	chip	G (0; 36%)	moderate	CTLA4: T lymphocyte brake. Meaning: GWAS Catalog: Alopecia areata (Betz RC 2015; p=2e-20); G ↑ allele

Typical / intermediate variants (4)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
RAET1M	rs12183587 6:150033472 · T/C/G	GT	chip	T (1; 56%)	moderate	ULBP/RAET1: NKG2D ligands expressed in the attacked follicle. Meaning: GWAS Catalog: Alopecia areata (Betz RC 2015; p=6e-24); T ↑ chip allele per position (ID rs9479478)
IL2RA	rs3118470 10:6059750 · T/A/C	TC	chip	C (1; 32%)	moderate	IL2RA (CD25): regulatory T lymphocytes. Meaning: GWAS Catalog: Alopecia areata (Betz RC 2015; p=8e-21); allele C ↑
PRDX5	rs574087 11:64335476 · A/G	GA	chip	A (1; 62%)	moderate	PRDX5: peroxiredoxin (oxidant stress in the follicle). Meaning: GWAS Catalog: Alopecia areata (Betz RC 2015; p=9e-14); allele A ↑
IKZF4	rs1701704 12:56018703 · T/A/G	TG	chip	G (1; 33%)	moderate	IKZF4 (Eos): regulatory T lymphocytes. Meaning: GWAS Catalog: Alopecia areata (Petukhova L 2010; p=3e-08); G ↑ allele

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HLA-DQB1	— · —	—	—	—	moderate	HLA class II region: the strongest locus of alopecia areata. No major variant of GWAS

SNP not
found in the
data

Catalog for this gene with usable genotype and
sense.

Hair nutrients and pharmacogenetics of treatments

Iron, vitamin D, folate/B12, biotin and zinc; and genes that change the response to minoxidil (SULT1A1, the follicle enzyme that activates it) and exposure to finasteride and dutasteride (CYP3A4, SRD5A2).

Ageing mechanisms: Metabolic Homeostasis decline; Endocrine dysregulation

Favourable

- HFE rs1800562 GG
- BTD rs13078881 GG

Unfavourable

- HFE rs1799945 CG

Favourable variants (2)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HFE	rs1800562 6:26092913 · G/A	GG	chip	A (0; 4%)	strong	C282Y: A allele is the main hereditary hemochromatosis (AA = iron overload).
BTB	rs13078881 3:15645186 · G/C/T	GG	chip	C (0; 4%)	weak	D444H (biotinidase): partial deficiency variant; only has a clinical effect in combination with another severe mutation of BTB.

Unfavourable variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
HFE	rs1799945 6:26090951 · C/G/T	CG	chip	G (1; 17%)	moderate	H63D: the G allele slightly increases iron absorption; the plan already fixed ferritin below 100.

Typical / intermediate variants (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
MTHFR	rs1801133 1:11796321 · G/A/C	GA	chip	A (1; 36%)	moderate	C677T: allele A (T in gene) reduces the activity of MTHFR and rises homocysteine; active folate and B12 cover this pathway.

Non-judgmental variants (informative) (3)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
TMPS6	rs855791 22:37066896 · A/C/G/T	AG	chip	—	moderate	V736A (matriptase-2, regulates hepcidin): an allele is associated with slightly lower iron and hemoglobin. Information.
FUT2	rs601338 19:48703417 · G/A	AG	chip	—	moderate	W154X: the allele A (non-secretory) is associated with higher serum B12. Information.
CYP3A4	rs35599367 7:99768693 · G/A	GG	chip	—	moderate	CYP3A4*22: less activity of CYP3A4, which metabolizes finasteride, dutasteride and rapamycin. Information.

SNP not found in the data (1)

Gene	Variant	Genotype	Source	Unfavourable allele (copying; frequency in reference)	Evidence	Interpretation
SULT1A1	rs9282861 — · —	—	—	—	weak	R213H (SULT1A1*2): less sulfotransferase activity that activates minoxidil in the follicle;

SNP not
found in
the data

the activity of SULT1A1 predicts the response
to minoxidil (the binding to the genotype is
indirect).

Mechanisms of ageing

Linking the results to the aging mechanisms of the list of objectives for immortality. Sorted by the number of unfavourable results.

Mechanism	Unfavourable	Favourable	Typical
Endocrine dysregulation	FAM53B CC; HFE CG	HDAC9 CC; TWIST2 CC; IRF4 AA; HFE GG; BTD GG	Baldness pattern (self-somes) (the 59th percentile); Male pattern (GWAS loci, autosomes) (the 32nd percentile); AR G; AR C; OPHN1 G; LINC01432 CG; EBF1 GA; SRD5A2 CC
Chronic Inflammation	FAM53B CC; Hair and hair follicle diseases (the 78th percentile)	HDAC9 CC; TWIST2 CC; IRF4 AA; CTLA4 AA	Baldness pattern (self-somes) (the 59th percentile); Male pattern (GWAS loci, autosomes) (the 32nd percentile); AR G; AR C; OPHN1 G; LINC01432 CG; EBF1 GA; SRD5A2 CC
Intercellular Communication	Hair and hair follicle diseases (the 78th percentile)	CTLA4 AA	Alopecia areata (the 28th percentile); RAET1M GT; IL2RA TC; PRDX5 GA; IKZF4 TG
Metabolic Homeostasis Decline	HFE CG	HFE GG; BTD GG	MTHFR GA
Exhaustion of Stem Cells	FAM53B CC	HDAC9 CC; TWIST2 CC; IRF4 AA	Baldness pattern (self-somes) (the 59th percentile); Male pattern (GWAS loci, autosomes) (the 32nd percentile); AR G; AR C; OPHN1 G; LINC01432 CG; EBF1 GA; SRD5A2 CC
Immunosenescence	Hair and hair follicle diseases (the 78th percentile)	CTLA4 AA	Alopecia areata (the 28th percentile); RAET1M GT; IL2RA TC; PRDX5 GA; IKZF4 TG
Cell Senescence	FAM53B CC	HDAC9 CC; TWIST2 CC; IRF4 AA	Baldness pattern (self-somes) (the 59th percentile); Male pattern (GWAS loci, autosomes) (the 32nd percentile); AR G; AR C; OPHN1 G; LINC01432 CG; EBF1 GA; SRD5A2 CC
Oxidative stress	—	—	IRF4 CC

Practical part

Ideas to protect hair, crossed with the genetic plan v2 and current medication (isotretinoin 40 mg/week, rapamycin, lithium 5 mg, tadalafil, dapagliflozin, retatrutide, telmisartan, LDN, melatonin). The current stack remains as it is; everything is suggestion (not applied). In androgenetic alopecia, the effect of DHT on the follicle and what prolongs the growth phase is more effective if it starts early.

1. Standardized photographs every 6 months

Genetics: Baldness pattern; AR in chromosome X

Photographs of the vertex, scratch and entrances with the same light and dry hair, every 6 months (or trichoscopy in dermatologist). The decision to treat should be based on real evolution and not only on genes.

V2 genetic plan and medication: New; no cost. • suggestion (not applied)

2. Ketoconazole 2% (double benefit)

Genetics: Baldness (mild local antiandrogenic effect); seborrheic dermatitis (CARD9 TT, see Skin report)

2–3×/week, leaving 3–5 min. In small studies it increased the density in a way comparable to minoxidil 2%, and treats Malassezia. To change with zinc pyrithione to dry.

V2 genetic plan and medication: Similar to the v2 plan (topic antifungals). • suggestion (not applied)

3. Topical minoxidil 5% (spume)

Genetics: Baldness

1 mL or half foam cover 1–2×/day on the dry scalp; results at 4–6 months, with transient fall in the first weeks. It only remains as long as it is used. To prefer the topic to oral: oral minoxidil in low dose is vasodilator and adds to telmisartan, tadalafil and dapagliflozin (twilights, fluid retention, tachycardia), and can cause hair in the body.

V2 genetic plan and medication: New; oral only with the doctor. • suggestion (not applied)

4. DHT blockage: topical or oral finasteride, or dutasteride (medical decision)

Genetics: SRD5A2 and AR (androgen)

Finasteride 1 mg oral (or topical spray 0.25%, with less systemic exposure) stops the baldness in about 85% of cases. Points to weigh: it reduces PSA to half (multiply by 2 when interpreting); possible sexual and mood effects, to take into account isotretinoin and lithium (watching mood, especially in the first months). The dutasteride is more potent, but the half-life is long.

V2 genetic plan and medication: New; requires prescription and discussion with the doctor. • suggestion (not applied)

5. Microneedling 1/week

Genetics: Baldness

Microneedling of 1.0–1.5 mm, 1×/week, combined with minoxidil (apply the minoxidil 24 h later, not on the same day). In trials increased the response to minoxidil. Do not do it while taking isotretinoin or in the 2–3 days after taking rapamycin (slower healing).

V2 genetic plan and medication: New; after isotretinoin. • suggestion (not applied)

6. Low intensity laser (LLLT) or PRP

Genetics: Baldness

Capacette or LLLT band (650–680 nm), 3×/week, with moderate and low risk evidence. Platelet-rich plasma (PRP), 3 monthly sessions and then maintenance, has moderate evidence; it is a procedure, so it should be away from taking rapamycin.

V2 genetic plan and medication: New; optional. • suggestion (not applied)

7. Iron: ferritin between 40 and 100

Genetics: HFE H63D heterozygous (trend to accumulate iron); TMPRSS6

The low ferritin (<30–40) aggravates the diffuse shedding; the high is pro-oxidant. Target of the plan: ferritin below 100; for hair above 40. Only supplement iron if it is below.

V2 genetic plan and medication: Coherent with the v2 plan (ferritin <100). • suggestion (not applied)

8. Protein and micronutrients already in the stack

Genetics: Diffuse shedding (telegenous efflux)

Protein of about 140 g/day, zinc + copper, vitamin D3 10 000 IU, omega-3, collagen 20 g and vitamin C already cover what the follicle needs. Biotin is not required without deficit and interferes with analysis (troponine, thyroid); if taken, stop 3 days before harvest.

V2 genetic plan and medication: Similar to the v2 plan. • [suggestion \(not applied\)](#)

9. Diffuse fall linked to medication

Genetics: — (drug effect, not genes)

Isotretinoin can cause telogene efflux; retatrutide can cause a decrease in 2–4 months after a rapid caloric deficit (the main factor is the deficit, not the drug), mitigated by the high protein; lithium can cause a drop in psychiatric doses (not likely with 5 mg). A sudden diffuse fall, without pattern, usually occurs in 6–12 months.

V2 genetic plan and medication: Background. • [suggestion \(not applied\)](#)

10. White hair

Genetics: IRF4 CC (without early greying allele)

There is no intervention with evidence to stop white hair; the state of copper, vitamin B12 and thyroid is already covered by analysis and stack.

V2 genetic plan and medication: Background. • [suggestion \(not applied\)](#)

Avoiding or with caution

- Oral minoxidil without the doctor (vasodilation added to telmisartan, tadalafil and dapagliflozin). [suggestion \(not applied\)](#)
- Microneedling, PRP or capillary transplantation during isotretinoin or with rapamycin. [suggestion \(not applied\)](#)
- High-dose biotin without deficit (interferes with analysis). [suggestion \(not applied\)](#)
- Penned with traction and excessive heat in the hair. [suggestion \(not applied\)](#)
- Supplements "for the hair" with vitamin A or retinol (they are added to isotretinoin). [suggestion \(not applied\)](#)

Limitations

- Genotyping chip + local imputation (public reference panel); imputed with $DR2 < 0.8$ treated as not found.
- The X chromosome (AR/EDA2R) explains a large part of the baldness and does not enter the autosomal scores; it is read by the main variants. The CAG/GGC repetitions of AR are not visible.
- The polygenic scores of baldness predict well the extremes but not the age of onset or speed; family history (especially on the maternal side) and direct observation are worth more.
- The binding between the SULT1A1 genotype and the minoxidil response is indirect (the studies measure the activity of the enzyme in the follicle, not the genotype).
- White hair has few known loci; IRF4 explains a small part.
- Percentages against 633 people from the reference population.

Methods and sources

Genotypes: first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local statistical imputation (with a public reference panel of complete genomes), only with $DR2 \geq 0.8$. The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

Variation classification: The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having ≥ 1 = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

Main variants per gene: for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ($p \leq 5 \times 10^{-8}$) is used, mapped to this gene and with available genotype.

Polygenic scores (PGS Catalog): Harmonized coefficients for GRCh38 applied to the imputed genome (chip + $DR2 \geq 0.8$, dosage). Empirical percentile in the reference population ($n=633$), calculated with the same variants. List: PGS001987 (PGS Catalog; Penalized regression; Hair/balding pattern; PGP000263, PMID 34995502 · PGS001873 (PGS Catalog; Penalized regression; Diseases of hair and hair follicles; PGP000263, PMID 34995502).

GWAS Catalog loci scores: for traits without PGS, weighted sum of risk variants ($p \leq 5 \times 10^{-8}$, one per window of 500kb), with a percentile calculated in the same way.

Percentiles: ≤ 25 or ≥ 75 (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

Main sources

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